Gomoku AI
Search Agents & Performance Analysis

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Game rules

Board size: 15×15

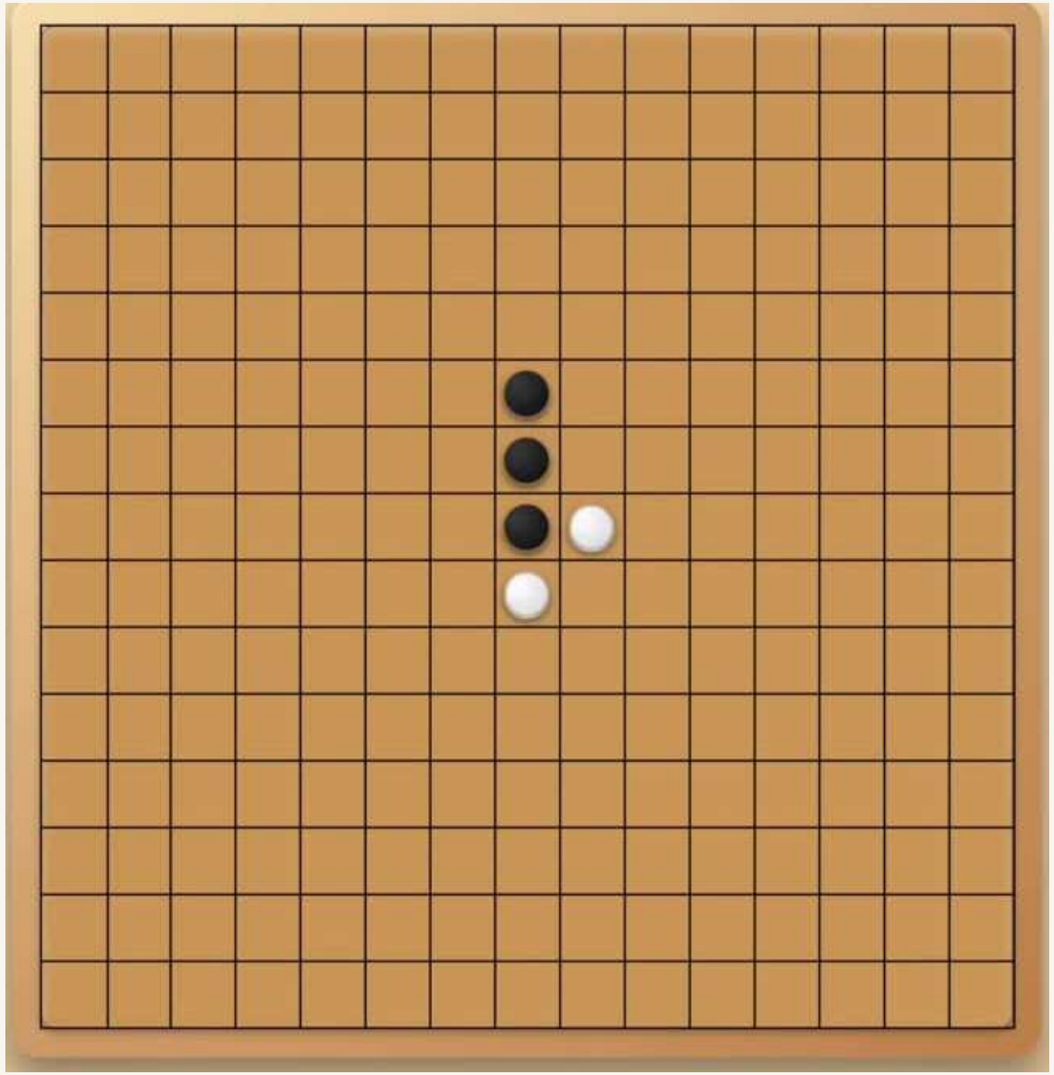
Players: Black and White

Turn order: Black moves first

Win condition: exactly 5 consecutive stones

Game ends when:

- One player gets 5 in a row
- Board is full → draw



Project Goal

Build AI agents capable of playing Gomoku on a 15×15 board.

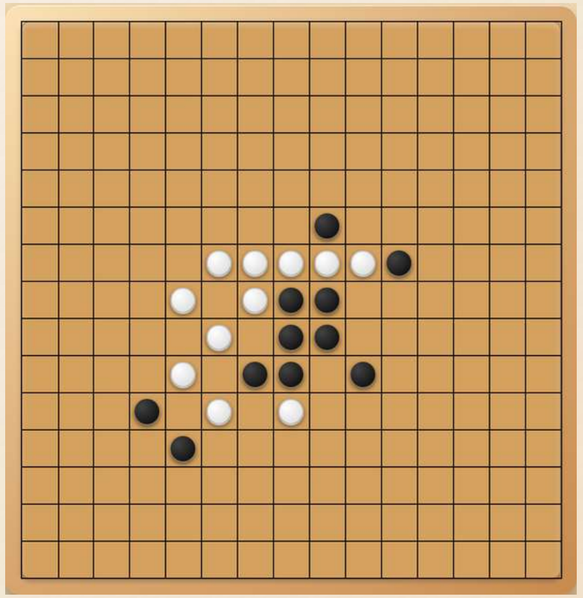
Compare their strength, efficiency, and scalability.

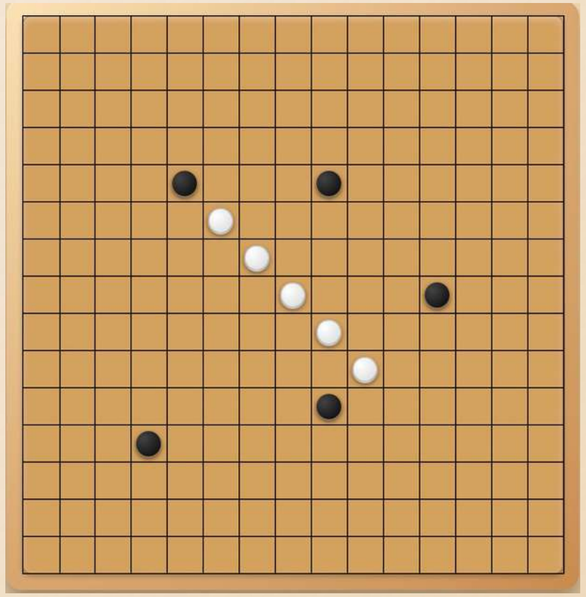
Determine how search depth and pruning influence performance.

Move Filtering Mechanism

Move Filtering (Proximity = 2)
Instead of evaluating all empty cells (225 possible), we consider only moves near existing stones. That:

- Reduces branching factor dramatically
- Enables deeper search (Depth 2–3)
- Still covers all meaningful moves





Branching Factor

Before Move Filtering

More than 220 legal moves early game and 200+ In mid-game

Most moves are strategically irrelevant

Search tree grows explosively:

Depth 2 → ~40,000 nodes

Depth 3 → ~8,000,000 nodes

After Move Filtering

After Move Filtering (k = 2 proximity)

Only consider moves near existing stones

Effective branching factor drops to 20–40 moves

Search tree becomes manageable:

Depth 2 → ~625 nodes

Depth 3 → ~15,625 nodes

Evaluation Function

Our evaluation function detects patterns for each player:

Winning sequences (five-in-a-row) **XXXXX (100,000)**
Open Four **_XXXX_ (50,000)**
Broken Four **_XX_XX_ or _XXX_X_ or _X_XXX_ (10,000)**
Closed Four **OXXXX_ or _XXXXO (3,000)**
Broken Three **_X_XX_ or _XX_X_ (2,000)**
Open Three **_XXX_ (800)**
Closed Three **OXXX_ or _XXXO (200)**

Total score = (Your patterns) – (Opponent's patterns)

Our Agents

Random Agent

Minimax Agent (Depth 1 & 2)

Alpha-Beta Agent (Depth 1, 2, 3)

Allow controlled comparison of: Search depth, Pruning efficiency. Computation vs. playing strengt

All agents share a similar interface: they receive the current board and must output a move based on either heuristic or randomness .

AlphaBeta Depth 2 VS Minimax Depth 2

Average total thinking time per game

- AlphaBeta(d=2): ≈ 22.3 s
- Minimax(d=2): ≈ 124.8 s

➔ Minimax takes about 5.6× more time.

Average nodes expanded per game

- AlphaBeta(d=2): ≈ 25,349 nodes
- Minimax(d=2): ≈ 136,696 nodes

➔ Minimax explores about 5.4× more nodes.

- **Average cutoffs for AlphaBeta**: ≈ 1,494.5 per game

Win Rate

AlphaBeta as first player: 60.0 %

Minimax as second player: 40.0 %

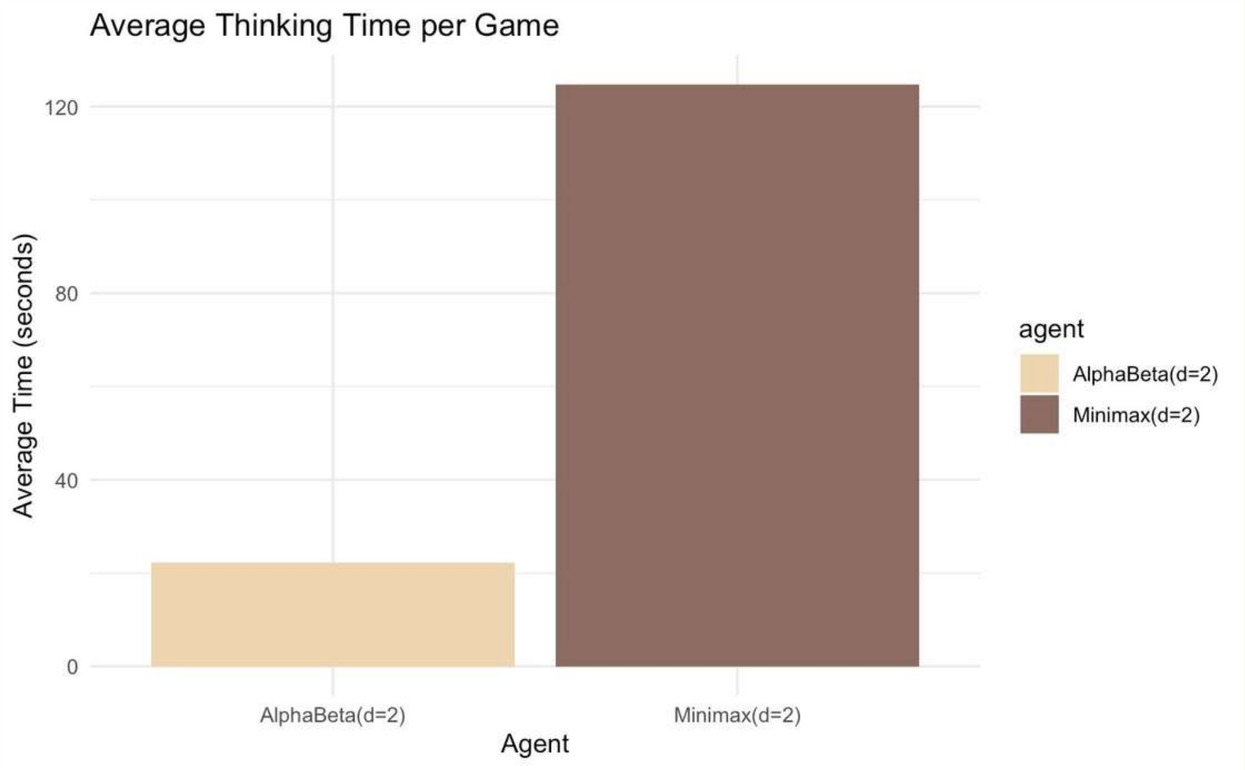
Draw 0 %

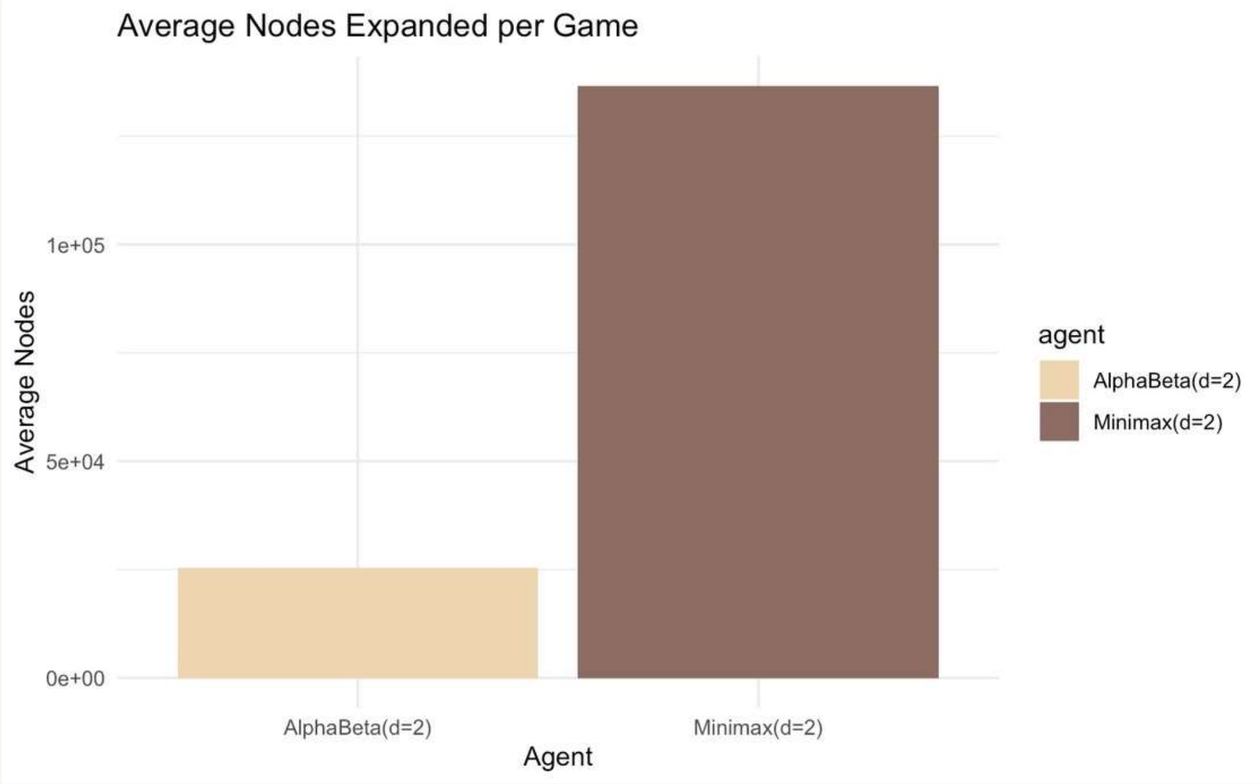
Minimax as first player: 73.3 %

AlphaBeta as second player: 20.0 %

Draw: 6.7 %

AlphaBeta Depth 2 VS Minimax Depth 2





AlphaBeta Depth 2 VS Minimax Depth 1

Average total thinking time per game

- AlphaBeta(d=2): ≈ 13.096s
- Minimax(d=1): ≈ 1.1s

➔ Minimax takes about 12× less time.

Average nodes expanded per game

- AlphaBeta(d=2): ≈ 9,968 nodes
- Minimax(d=2): ≈ 811 nodes

➔ Minimax explores about 5.4× less nodes.

Average cutoffs for AlphaBeta: ≈ 819 per game

Win Rate

AlphaBeta(d=2) as first player: 100.0 %

Minimax(d=1) as second player: 0.0 %

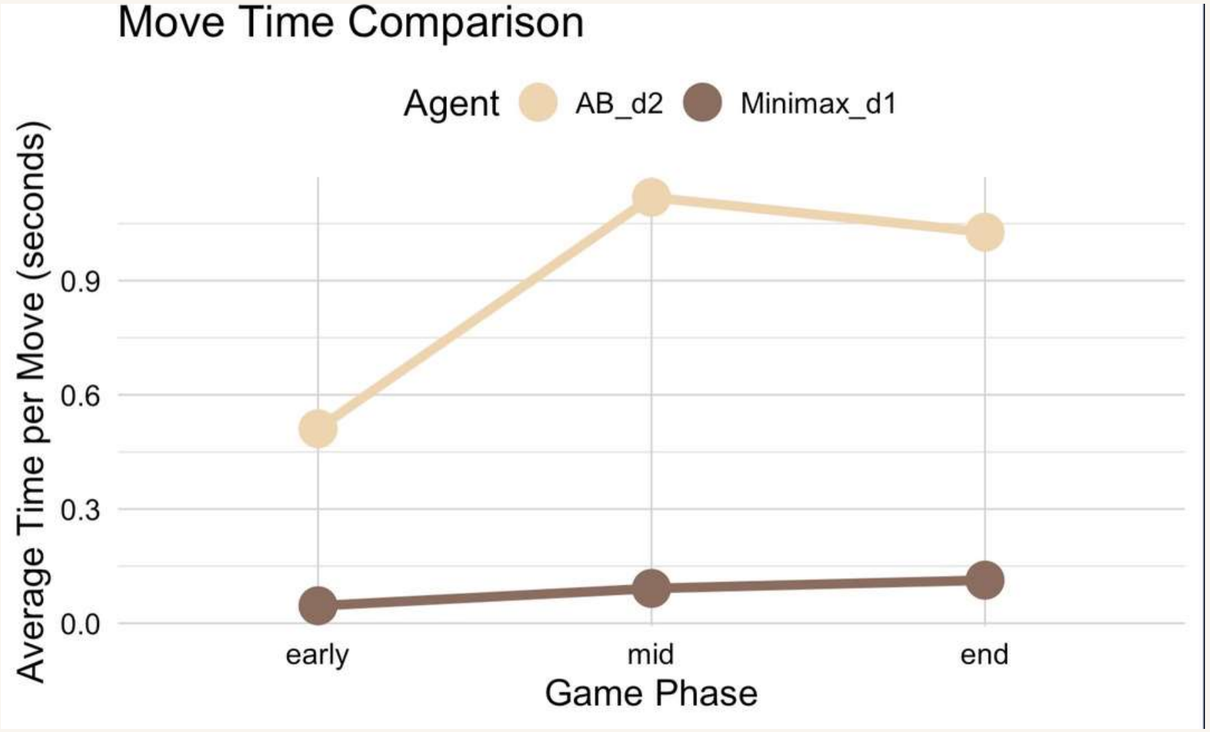
Draw: 0.0 %

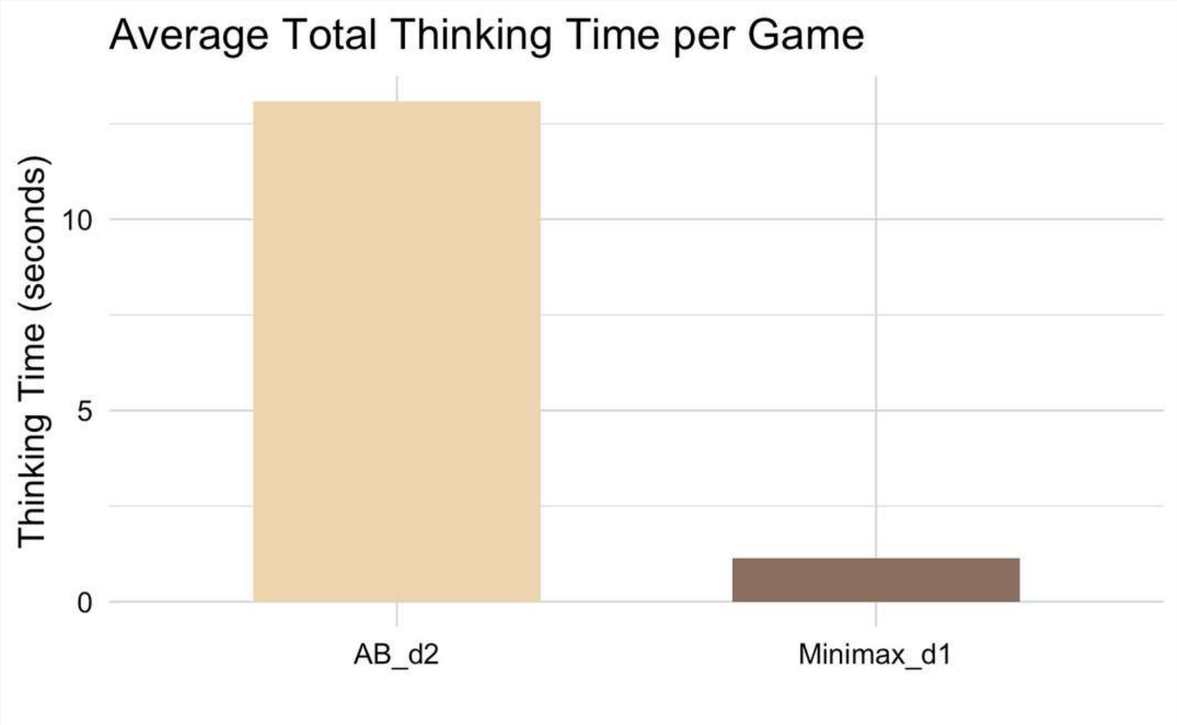
Minimax(d=1) as first player: 6.7 %

AlphaBeta(d=2) as second player: 93.3 %

Draw: 0.0 %

Thinking Time





Search Agent VS Random Agent

Average total thinking time per game

- Search Agent (d=2 or d=1): ≈ 2.25
- Random: ~0 s (negligible thinking time)

Average nodes expanded per game

- Search Agent: expands thousands of nodes per game
- Random: expands almost none (0–5 nodes)

Win Rate (vs Random)

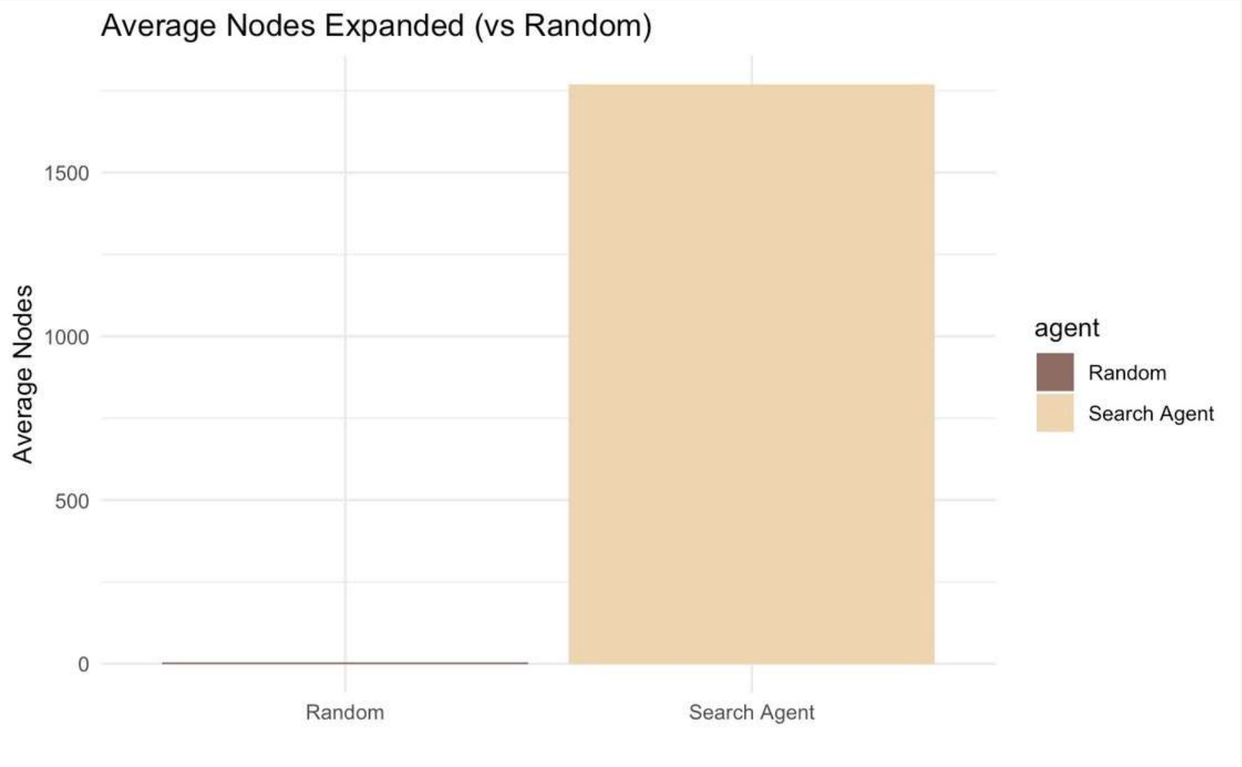
AlphaBeta(d=2) vs Random

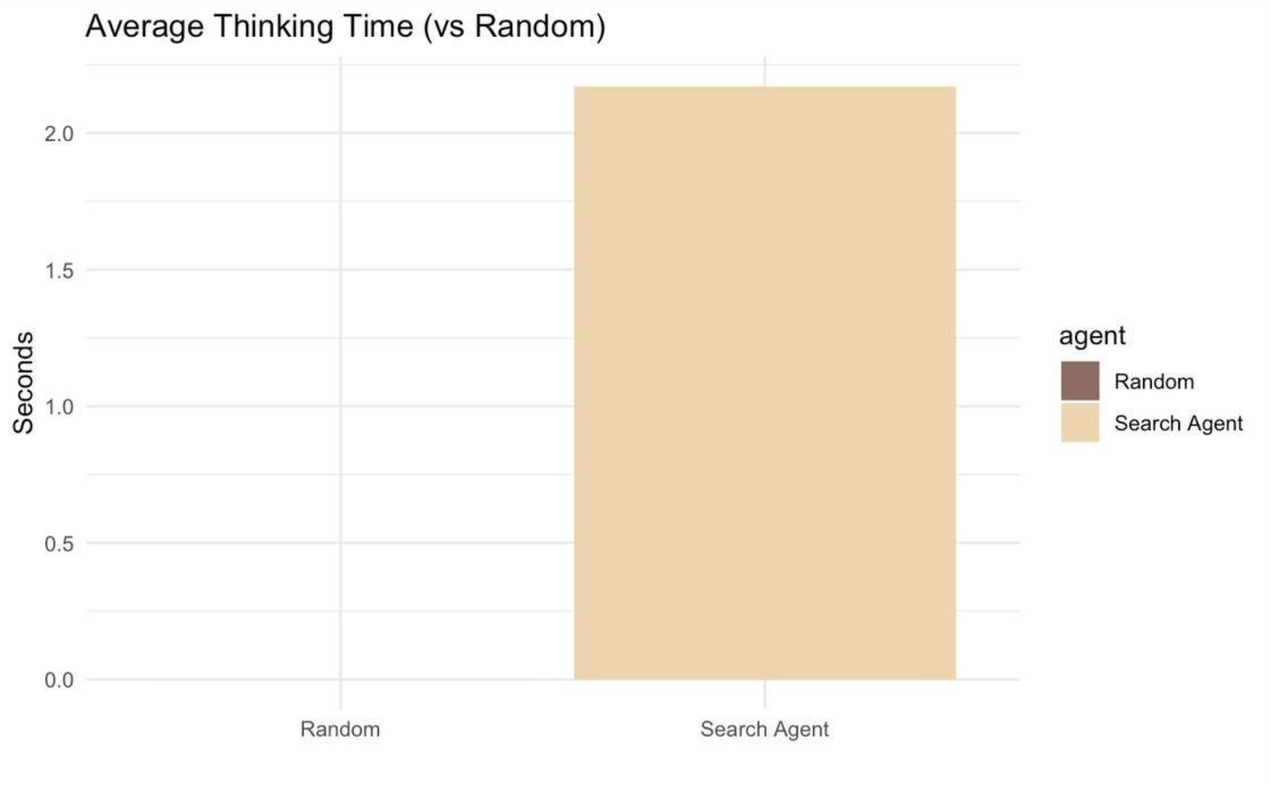
- AlphaBeta wins: 100%
- Random wins: 0%
- Draw: 0%

Minimax(d=1) vs Random

- Minimax wins: 100%
- Random wins: 0%
- Draw: 0%

Search Agent VS Random Agent





AlphaBeta Depth 2 VS AlphaBeta Depth 3

Average total thinking time per game

- AlphaBeta(d=2): ≈ 17.89 s
- AlphaBeta(d=3): ≈ 303.93 s
- ➔ Depth 3 takes ~17× more time per game.

Average nodes expanded per game

- AlphaBeta(d=2): ≈ 20,740 nodes
- AlphaBeta(d=3): ≈ 339,962 nodes
- ➔ Depth 3 explores ~16.4× more nodes

Average cutoffs

- AlphaBeta(d=2): ≈ 805 cutoffs
- AlphaBeta(d=3): ≈ 7,555 cutoffs
- ➔ Depth 3 produces ~9.4× more cutoffs

Win Rate

AlphaBeta(d=2) as first player: 37.5 %

AlphaBeta(d=3) as second player: 62.5 %

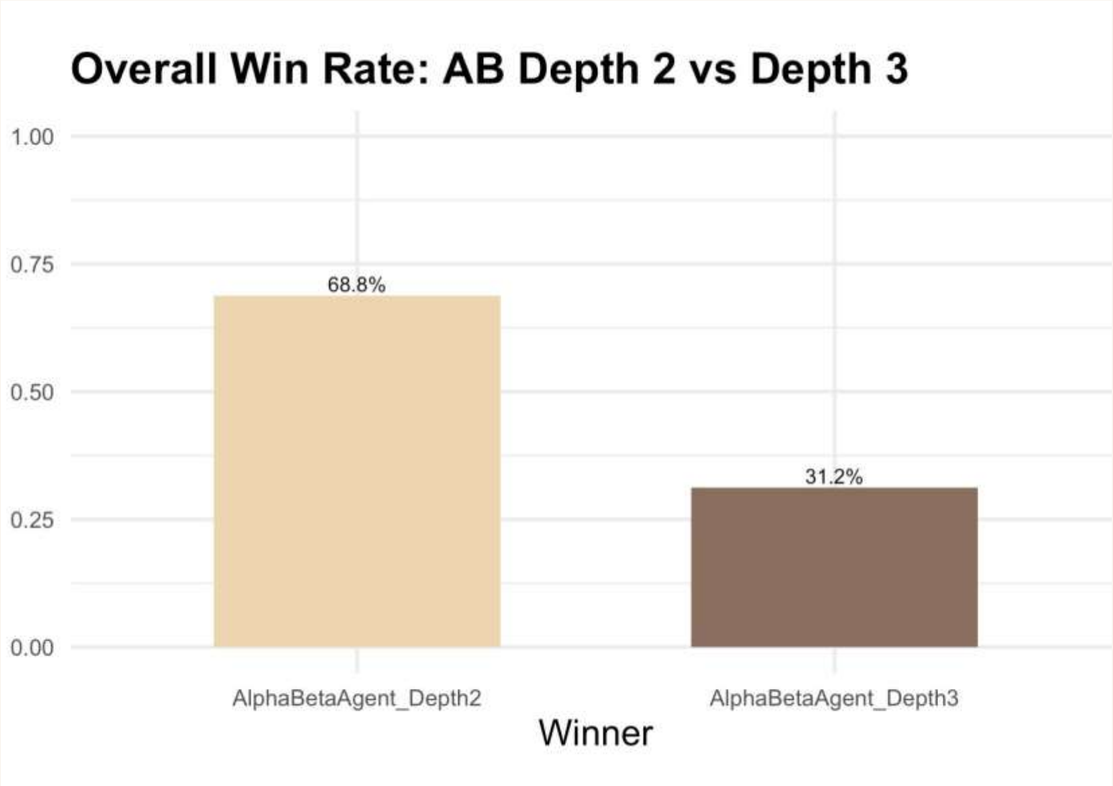
Draw: 0.0 %

AlphaBeta(d=3) as first player: 0.0 %

AlphaBeta(d=2) as second player: 100.0 %

Draw: 0.0 %

Alpha-Beta Depth 2 vs Depth 3



Despite deeper search, Depth3 is less effective overall, while Depth2 demonstrates stronger tactical consistency and completely dominates when responding second.

Thank You

Questions?